

Breadboard 8088 PC Ver 5.0

<table><tr><td>GND</td><td>VCC</td></tr><tr><td>A14</td><td>A15</td></tr><tr><td>A13</td><td>A16S3</td></tr><tr><td>A12</td><td>A17S4</td></tr><tr><td>A11</td><td>A18S5</td></tr><tr><td>A10</td><td>A19S6</td></tr><tr><td>A9</td><td>HIGH</td></tr><tr><td>A8</td><td>MNMX</td></tr><tr><td>AD7</td><td>P8088 RD</td></tr><tr><td>AD6</td><td>RQGT0</td></tr><tr><td>AD5</td><td>RQGT1</td></tr><tr><td>AD4</td><td>TDCK</td></tr><tr><td>AD3</td><td>MSZ</td></tr><tr><td>AD2</td><td>ST</td></tr><tr><td>AD1</td><td>S0</td></tr><tr><td>AD0</td><td>QS0</td></tr><tr><td>NMI</td><td>QS1</td></tr><tr><td>INTR</td><td>TEST</td></tr><tr><td>CLK</td><td>READY</td></tr><tr><td>GND</td><td>RESET</td></tr></table>	GND	VCC	A14	A15	A13	A16S3	A12	A17S4	A11	A18S5	A10	A19S6	A9	HIGH	A8	MNMX	AD7	P8088 RD	AD6	RQGT0	AD5	RQGT1	AD4	TDCK	AD3	MSZ	AD2	ST	AD1	S0	AD0	QS0	NMI	QS1	INTR	TEST	CLK	READY	GND	RESET	<table><tr><td>CsycgVcc</td><td>A7</td></tr><tr><td>Pclk 2x18</td><td>A6</td></tr><tr><td>Aen14X2</td><td>A5</td></tr><tr><td>Rdy1X5y</td><td>A4</td></tr><tr><td>Rdy EF1</td><td>A3</td></tr><tr><td>Rdy2 F/C</td><td>A2</td></tr><tr><td>Aen2 Osc</td><td>A1</td></tr><tr><td>CLK Res</td><td>A0</td></tr><tr><td>GND Rst</td><td></td></tr></table> <table><tr><td>IOB8/Vcc</td><td>S0</td></tr><tr><td>CLK S0</td><td>S1</td></tr><tr><td>ST S2</td><td>S3</td></tr><tr><td>Dt/Kmpd</td><td>S4</td></tr><tr><td>ALE Den</td><td>S5</td></tr><tr><td>AEN Cen</td><td>S6</td></tr><tr><td>mrd Tnta</td><td>S7</td></tr><tr><td>amwTofc</td><td>S8</td></tr><tr><td>mwTAtow</td><td>S9</td></tr><tr><td>GNDTow</td><td></td></tr></table>	CsycgVcc	A7	Pclk 2x18	A6	Aen14X2	A5	Rdy1X5y	A4	Rdy EF1	A3	Rdy2 F/C	A2	Aen2 Osc	A1	CLK Res	A0	GND Rst		IOB8/Vcc	S0	CLK S0	S1	ST S2	S3	Dt/Kmpd	S4	ALE Den	S5	AEN Cen	S6	mrd Tnta	S7	amwTofc	S8	mwTAtow	S9	GNDTow		<table><tr><td>TOR</td><td>A7</td></tr><tr><td>TOW</td><td>A6</td></tr><tr><td>MEMR</td><td>A5</td></tr><tr><td>MEMW</td><td>A4</td></tr><tr><td>Pin 5</td><td>A3</td></tr><tr><td>Ready</td><td>A2</td></tr><tr><td>HLDA</td><td>A1</td></tr><tr><td>ADstb</td><td>A0</td></tr><tr><td>AEN</td><td>A0</td></tr><tr><td>HRQ</td><td>VCC</td></tr><tr><td>CS</td><td>DB0</td></tr><tr><td>CLK</td><td>DB1</td></tr><tr><td>RESET</td><td>DB2</td></tr><tr><td>Dack2</td><td>DB3</td></tr><tr><td>Dack3</td><td>DB4</td></tr><tr><td>Dreq3</td><td>Dack0</td></tr><tr><td>Dreq2</td><td>Dack1</td></tr><tr><td>Dreq1</td><td>DB5</td></tr><tr><td>Dreq0</td><td>DB6</td></tr><tr><td>GND</td><td>DB7</td></tr></table>	TOR	A7	TOW	A6	MEMR	A5	MEMW	A4	Pin 5	A3	Ready	A2	HLDA	A1	ADstb	A0	AEN	A0	HRQ	VCC	CS	DB0	CLK	DB1	RESET	DB2	Dack2	DB3	Dack3	DB4	Dreq3	Dack0	Dreq2	Dack1	Dreq1	DB5	Dreq0	DB6	GND	DB7	<table><tr><td>D2</td><td>8</td><td>D1</td></tr><tr><td>D3</td><td>2</td><td>D0</td></tr><tr><td>RxD</td><td>1</td><td>Vcc</td></tr><tr><td>GND</td><td></td><td>RxC</td></tr><tr><td>D4</td><td>U</td><td>DTR</td></tr><tr><td>D5</td><td>A</td><td>RTS</td></tr><tr><td>D6</td><td>D</td><td>DSR</td></tr><tr><td>D7</td><td>R</td><td>RESET</td></tr><tr><td>TxC</td><td></td><td>CLK</td></tr><tr><td>WR</td><td></td><td>TxD</td></tr><tr><td>CS</td><td></td><td>TxEmp</td></tr><tr><td>C/D</td><td></td><td>CTS</td></tr><tr><td>RD</td><td></td><td>Syn/Bd</td></tr><tr><td>RxRdy</td><td></td><td>TxRdy</td></tr></table>	D2	8	D1	D3	2	D0	RxD	1	Vcc	GND		RxC	D4	U	DTR	D5	A	RTS	D6	D	DSR	D7	R	RESET	TxC		CLK	WR		TxD	CS		TxEmp	C/D		CTS	RD		Syn/Bd	RxRdy		TxRdy	<table><tr><td>D7</td><td>Vcc</td></tr><tr><td>D6</td><td>WR</td></tr><tr><td>D5</td><td>RD</td></tr><tr><td>D4</td><td>CS</td></tr><tr><td>D3</td><td>A1</td></tr><tr><td>D2</td><td>A0</td></tr><tr><td>D1</td><td>CLK2</td></tr><tr><td>D0</td><td>A0 OUT2</td></tr><tr><td>CLK0</td><td>Gate2</td></tr><tr><td>OUT0</td><td>CLK1</td></tr><tr><td>Gate0</td><td>Gate1</td></tr><tr><td>GND</td><td>OUT1</td></tr></table>	D7	Vcc	D6	WR	D5	RD	D4	CS	D3	A1	D2	A0	D1	CLK2	D0	A0 OUT2	CLK0	Gate2	OUT0	CLK1	Gate0	Gate1	GND	OUT1	<table><tr><td>CS</td><td>8</td><td>Vcc</td></tr><tr><td>WR</td><td>2</td><td>A0</td></tr><tr><td>RD</td><td>5</td><td>INTA</td></tr><tr><td>D7</td><td>IR7</td></tr><tr><td>D6</td><td>I</td><td>IR6</td></tr><tr><td>D5</td><td>S</td><td>IR5</td></tr><tr><td>D4</td><td>I</td><td>IR4</td></tr><tr><td>D3</td><td>C</td><td>IR3</td></tr><tr><td>D2</td><td>n</td><td>IR2</td></tr><tr><td>D1</td><td>t</td><td>IR1</td></tr><tr><td>D0</td><td>r</td><td>IR0</td></tr><tr><td>CAS0</td><td>I</td><td>INT</td></tr><tr><td>CAS1</td><td>e</td><td>SP/EN</td></tr><tr><td>GND</td><td>r</td><td>CAS2</td></tr></table>	CS	8	Vcc	WR	2	A0	RD	5	INTA	D7	IR7	D6	I	IR6	D5	S	IR5	D4	I	IR4	D3	C	IR3	D2	n	IR2	D1	t	IR1	D0	r	IR0	CAS0	I	INT	CAS1	e	SP/EN	GND	r	CAS2	<table><tr><td>PA3</td><td>PA4</td></tr><tr><td>PA2</td><td>8</td><td>PA5</td></tr><tr><td>PA1</td><td>2</td><td>PA6</td></tr><tr><td>PA0</td><td>5</td><td>PA7</td></tr><tr><td>RD</td><td>p</td><td>WR</td></tr><tr><td>CS</td><td>e</td><td>RESET</td></tr><tr><td>GND</td><td>i</td><td>D0</td></tr><tr><td>A1</td><td>D1</td><td>D1</td></tr><tr><td>A0</td><td>D2</td><td>D2</td></tr><tr><td>PC7</td><td>D3</td><td>D3</td></tr><tr><td>PC6</td><td>D4</td><td>D4</td></tr><tr><td>PC5</td><td>D5</td><td>D5</td></tr><tr><td>PC4</td><td>D6</td><td>D6</td></tr><tr><td>PC0</td><td>D7</td><td>D7</td></tr><tr><td>PC1</td><td>Vcc</td><td></td></tr><tr><td>PC2</td><td>PB7</td><td></td></tr><tr><td>PC3</td><td>PB6</td><td></td></tr><tr><td>PB0</td><td>PB5</td><td></td></tr><tr><td>PB1</td><td>PB4</td><td></td></tr><tr><td>PB2</td><td>PB3</td><td></td></tr></table>	PA3	PA4	PA2	8	PA5	PA1	2	PA6	PA0	5	PA7	RD	p	WR	CS	e	RESET	GND	i	D0	A1	D1	D1	A0	D2	D2	PC7	D3	D3	PC6	D4	D4	PC5	D5	D5	PC4	D6	D6	PC0	D7	D7	PC1	Vcc		PC2	PB7		PC3	PB6		PB0	PB5		PB1	PB4		PB2	PB3		<table><tr><td>Vss</td><td>VS</td></tr><tr><td>RESET</td><td>HS</td></tr><tr><td>LPSTB</td><td>RA0</td></tr><tr><td>MA0</td><td>RA1</td></tr><tr><td>MA1</td><td>RA2</td></tr><tr><td>MA2</td><td>RA3</td></tr><tr><td>MA3</td><td>RA4</td></tr><tr><td>MA4</td><td>D0</td></tr><tr><td>MA5</td><td>D1</td></tr><tr><td>MA6</td><td>D2</td></tr><tr><td>MA7</td><td>D3</td></tr><tr><td>MA8</td><td>D4</td></tr><tr><td>MA9</td><td>D5</td></tr><tr><td>MA10</td><td>D6</td></tr><tr><td>MA11</td><td>D7</td></tr><tr><td>MA12</td><td>CS</td></tr><tr><td>MA13</td><td>RS</td></tr><tr><td>DE</td><td>E</td></tr><tr><td>CUR</td><td>R/W</td></tr><tr><td>Vcc</td><td>CLK</td></tr></table>	Vss	VS	RESET	HS	LPSTB	RA0	MA0	RA1	MA1	RA2	MA2	RA3	MA3	RA4	MA4	D0	MA5	D1	MA6	D2	MA7	D3	MA8	D4	MA9	D5	MA10	D6	MA11	D7	MA12	CS	MA13	RS	DE	E	CUR	R/W	Vcc	CLK	<table><tr><td>TEST0</td><td>Vcc</td></tr><tr><td>XTAL1</td><td>TEST1</td></tr><tr><td>XTAL2</td><td>P27</td></tr><tr><td>RESET</td><td>P26</td></tr><tr><td>THSS</td><td>P25</td></tr><tr><td>CS</td><td>P24</td></tr><tr><td>TLEA</td><td>P17</td></tr><tr><td>RD</td><td>P16</td></tr><tr><td>A0</td><td>P15</td></tr><tr><td>WR</td><td>P14</td></tr><tr><td>SYNC</td><td>P13</td></tr><tr><td>D0</td><td>P12</td></tr><tr><td>D1</td><td>P11</td></tr><tr><td>D2</td><td>P10</td></tr><tr><td>D3</td><td>THSSP</td></tr><tr><td>D4</td><td>THPR</td></tr><tr><td>D5</td><td>P23</td></tr><tr><td>D6</td><td>P23</td></tr><tr><td>D7</td><td>P21</td></tr><tr><td>GND</td><td>P20</td></tr></table>	TEST0	Vcc	XTAL1	TEST1	XTAL2	P27	RESET	P26	THSS	P25	CS	P24	TLEA	P17	RD	P16	A0	P15	WR	P14	SYNC	P13	D0	P12	D1	P11	D2	P10	D3	THSSP	D4	THPR	D5	P23	D6	P23	D7	P21	GND	P20	<table><tr><td>Red</td><td>Vdd</td></tr><tr><td>Green</td><td>RS1</td></tr><tr><td>Blue</td><td>RS0</td></tr><tr><td>IRef</td><td>WR</td></tr><tr><td>P0</td><td>G</td><td>D7</td></tr><tr><td>P1</td><td>1</td><td>D6</td></tr><tr><td>P2</td><td>6</td><td>D5</td></tr><tr><td>P3</td><td>P</td><td>D4</td></tr><tr><td>P4</td><td>D3</td><td></td></tr><tr><td>P5</td><td>D2</td><td></td></tr><tr><td>P6</td><td>D1</td><td></td></tr><tr><td>P7</td><td>D0</td><td></td></tr><tr><td>CLK</td><td>Blank</td><td></td></tr><tr><td>Vss</td><td>RD</td><td></td></tr></table>	Red	Vdd	Green	RS1	Blue	RS0	IRef	WR	P0	G	D7	P1	1	D6	P2	6	D5	P3	P	D4	P4	D3		P5	D2		P6	D1		P7	D0		CLK	Blank		Vss	RD		<table><tr><td>Vss</td><td>CA1</td></tr><tr><td>PA0</td><td>CA2</td></tr><tr><td>PA1</td><td>IRQA</td></tr><tr><td>PA2</td><td>IRQB</td></tr><tr><td>PA3</td><td>RS0</td></tr><tr><td>PA4</td><td>RS1</td></tr><tr><td>PA5</td><td>RESET</td></tr><tr><td>PA6</td><td>D0</td></tr><tr><td>PA7</td><td>M</td><td>D1</td></tr><tr><td>PB0</td><td>C</td><td>D2</td></tr><tr><td>PB1</td><td>8</td><td>D3</td></tr><tr><td>PB2</td><td>B</td><td>D4</td></tr><tr><td>PB3</td><td>2</td><td>D5</td></tr><tr><td>PB4</td><td>D6</td><td></td></tr><tr><td>PB5</td><td>D7</td><td></td></tr><tr><td>PB6</td><td>E</td><td></td></tr><tr><td>PB7</td><td>CS1</td><td></td></tr><tr><td>CB1</td><td>CS2</td><td></td></tr><tr><td>CB2</td><td>CS0</td><td></td></tr><tr><td>Vcc</td><td>R/W</td><td></td></tr></table>	Vss	CA1	PA0	CA2	PA1	IRQA	PA2	IRQB	PA3	RS0	PA4	RS1	PA5	RESET	PA6	D0	PA7	M	D1	PB0	C	D2	PB1	8	D3	PB2	B	D4	PB3	2	D5	PB4	D6		PB5	D7		PB6	E		PB7	CS1		CB1	CS2		CB2	CS0		Vcc	R/W	
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74 Series Logic Chips (0I indicate zero indexed)

Dir Vcc			Dir Vcc			G1 Vcc			G1 Vcc			OE Vcc			OE Vcc			OE Vcc			OE Vcc			OE Vcc			OE Vcc			CAB Vcc			CAB Vcc			E			A9			I/O5L I/O3R					
A0 7 OE			A0 7 OE			A6 7 G2			A0 7 G2			0D 7 Q0			0D 7 Q0			0D 7 Q0			0D 7 Q0			0D 7 Q0			0D 7 Q0			7D 7 Q0			SAB CBA			SAB CBA			RESET			A8			I/O6L I/O2R		
A1 4 B0			A1 4 B0			A4 4 Y7			A1 4 Y0			1D 4 A1Q			1D 4 A1Q			1D 4 A1Q			1D 4 A1Q			1D 4 A1Q			1D 4 A1Q			6D 4 A1Q			DIR SBA			DIR SBA						A7			I/O7L I/O1R		
A2 A B1			A2 A B1			A5 L Y6			A4 L Y1			2D L 2Q			2D L 2Q			2D L 2Q			2D L 2Q			2D L 2Q			2D L 2Q			5D L 5Q			A0 OE			D0 OE			A6								
A3 S B2			A3 S B2			A4 S Y5			A3 S Y2			3D S 3Q			3D S 3Q			3D S 3Q			3D S 3Q			3D S 3Q			3D S 3Q			4D 7 4Q			D1 1 id8			D1 1 id8											
A4 2 B3			A4 2 B3			A5 S Y4			A4 S Y3			4D 7 4Q			4D 7 4Q			4D 7 4Q			4D 7 4Q			4D 7 4Q			4D 7 4Q			5D 4 5Q			2D 7 3Q			D2 S id9			A5								
A5 4 B4			A5 4 B4			A2 1 Y3			A5 1 Y4			5D 3 5Q			5D 3 5Q			5D 4 5Q			5D 4 5Q			5D 4 5Q			5D 4 5Q			1D 0 1Q			3D 7 id10			D3 S id10			A4								
A6 B5			A6 B5			A1 0 Y2			A6 0 Y5			6D 0 6Q			6D 0 6Q			6D 0 6Q			6D 0 6Q			6D 0 6Q			6D 0 6Q			1D 0 1Q			A3 S B2			D4 6 id11			Q0								
A7 0 B6			A7 0 B6			A0 1 Y1			A7 R Y6			7D 1 7Q			7D 1 7Q			7D 1 7Q			7D 1 7Q			7D 1 7Q			7D 1 7Q			0D R 0Q			A4 4 B3			D5 1 id12			Q1								
GND B7			GND B7			GND Y0			GND Y7			GND LE			GND LE			GND CLK			GND CLK			GND CLK			GND CLK			GND CLK			A5 6 B4			D6 E id13			Q2								
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PLD Programmable Logic Devices

A19 Vcc	A19 Vcc	I019 Vcc	25Mz Vcc	AT7 Vcc	CCc Vcc	NC Vcc	Clk Vcc	25Mz Vcc	I0C1k Vcc	CK10 Vcc	PC1k Vcc	I019 Vcc	Out1 Vcc	CLK Vcc	MoteA	4	G
A18 I00	A18 I00	TOR CTrs	RA0 ATL	AT6 NO	Chc HS	PB3 A P3	V5InA V5o	RA0 Asrl	I1 Aio19	I1 I00	CK8 Apmc	A2 DTrw	Rst Arefl	Rst A DW	Y	5	D
A17 A101	A17 A10A	TOW 64S	RA1 URA	AT5 UNL	BuT VS	PB2 T P2	HSInT HSq	RA1 URA	I2 A101	I2 A101	RST Tpmc	A3 A dnm	SpkT I01	TOR T ABR	D	6	I
A16 F102	A16 F10A	A0 F CTrs	RA2 SRL	AT4 RVV	BuCF HSq	PB1 P1	InV5 NC	RA2 VAO	I3 F1017	I3 T102	I3 F1 CLK	A4 T P1r	Out2 I02	TOW F AEN	D	7	S
A15 CTrm1	A15 CTrm2	A1 2 NC	RA3 CHL	AT3 ADT	VE 2 V5d	PB0 P0	InH5 BH4	RA3 CHL	I4 61016	I4 1023	I4 6 CLK	A5 P1EL	I4 2103	I4 2 RRD	K	8	A
A14 VPIA	A14 VPIA	A2 V PRD	RA4 VPIA	AT2 VAD	g Sdu3	NC Q3	NC g Bu3	RST VQ	I5 V1015	I5 2104	I5 V CLK3	A6 2 PBF	I5 104	I5 14 PIA	K	9	P
A13 Tron	A13 Tron	A3 1 PWR	NC 1 Qb	AT1 P4	JUR4 NC	NC Q2	NC M Q2	NC 1 Qb	I6 B1014	I6 L105	I6 B CLK4	A7 V PWB	I6 1105	LOCK I05	K	10	C
A12 DTrm1	A12 DTrm	A4 1 NC	NC M CTrc	AT0 A3	B1EM NC	NC S1	NC B Q1	NC M CTrc	I7 1013	I7 1106	I7 S CLK5	A8 1 P1E2	I7 0106	SD 0 I06	D	11	Y
MinE E	Tor E E	A5 0 A11	NC C CTrc	VENF P2	NO D NC	NC E Q0	NC S Q0	NC C Qr	I8 1012	I8 0107	I8 CRST	A9 1 P0Sg	I8 p107	W1 WRW1	(F)	12	E
Minw RST	Tow RST	A6 V A10	NC 1 Dst	BW M P1	JVLA VID	GND P NC	GND 1 DE	NC V vad	I9 108	I9 108	GND DE	A101 Tsw	I9 B TC	Hrg D Trw	3	13	F
DEN IDEN	DEN 3DEN	A7 1 A9	NC 1 Dst	B1Ew P0	R1N1 VID		GND 2 Dot	NC 2 Dot	I10 I09	I10 I09	GND DE	A11 D A10	TC 0 S1P0	Holdg RD1	4	14	G
GND RST	GND RST	GND A8	GND NC	GND NC	GND AT7		GND NC	GND NC	GND 111	GND 111		GND A0	GND I10	GND Rdw			